

Clinical Metadata-Conditioned Query Formers for Volumetric Chest CT Understanding

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Abstract

Vision-language models for volumetric chest CT ordinarily infer abnormalities from image features and generic pathology prompts, although clinical indication and demographics are available at interpretation time and can substantially alter disease priors. We introduce METAQ-CT, a metadata-conditioned Q-Former built on an anatomy-routed 3-D CT encoder. The model retains a metadata-independent query bank while adding pathology-specific conditioned queries that attend to indication, age, and sex through gated cross-attention and identity-initialized feature-wise modulation. On a historical patient-disjoint age- ≥ 5 pediatric validation cohort with 23 findings, the proposed prompt readout improves three-seed macro AUROC from 0.7933 ± 0.0017 to 0.8396 ± 0.0015 (+4.63 points). Blanking or shuffling indications reduces AUROC to 0.8092 and 0.7960, respectively, demonstrating dependence on the correct patient context. On the official 18-label CT-RATE validation split, the same design improves the image-only prompt branch from 0.8733 to 0.8781 on one final checkpoint. The inherited general representation also reaches 13.07/20.29/44.88/57.27% report-to-CT recall at $K = 5/10/50/100$ without retrieval-specific fitting. These results show that patient-specific context is particularly valuable for pediatric CT while preserving strong adult classification and retrieval.

1 Introduction

Three-dimensional chest CT contains small, spatially localized abnormalities whose interpretation depends on both visual evidence and the clinical question that motivated the examination. Existing CT vision-language systems such as CT-CLIP [1], MPS-CT [3], GreenRFM [4], and ARC-CT [2] learn strong image-text representations, but their standard abnormality readouts are not explicitly conditioned on patient-specific indication and demographics.

This limitation is especially relevant in pediatric radiology. Indications often encode malignancy history, transplant status, postoperative context, or a focused diagnostic concern. A generic global metadata embedding, however, can impose the same correction on unrelated pathologies and allow helpful and harmful effects to cancel. We instead condition pathology-specific query slots while preserving a separate general representation.

Our contributions are:

- a metadata-conditioned Q-Former that combines clinical cross-attention, identity-initialized FiLM, and class-specific residual prompt corrections.
- a controlled three-seed pediatric evaluation with matched CT-only models, age-stratified and per-class analyses, and true/blank/shuffled indication interventions.
- a matched adult CT-RATE study showing that the architecture improves the classical 18-label benchmark. It also reveals a much smaller causal contribution from adult indication text.
- downstream report-to-CT retrieval and preliminary report-generation analyses that distinguish inherited representation quality from gains caused specifically by metadata conditioning.

2 Method

2.1 Base volumetric representation

Given a CT volume X , a 3-D ResNet encoder produces visual features that are processed by an anatomy-aware Q-Former initialized from ARC-CT. For C pathology targets, the general bank contains 10 anatomy queries, C pathology queries, and two global queries. Its pooled representation z_{gen} is deliberately isolated from metadata-conditioned slots, retaining a canonical image/report embedding for prompt inference and retrieval. The adult Biltir pipeline permits examinations without a segmentation mask, in which case attention is unrestricted. The primary frozen pediatric recipe requires the available TotalSegmentator masks. Mask-free operation is tested as a registered ablation rather than silently mixed into the primary arm.

2.2 Clinical context encoder

The clinical context combines indication text I , a learned scalar or banded age representation a , sex s , and an age-sex interaction:

$$H_C = \text{ContextEncoder}(I, a, s). \quad (1)$$

Missing indications are passed to the frozen clinical text tower as the literal string `NO_INDICATION`. They are never replaced with fabricated clinical text. Configuration is cohort specific: the current frozen pediatric primary

arm uses learned age bands and 30% indication dropout, whereas the historical local Refined-C1 checkpoints used scalar age and 10% dropout. The adult Biltir prompt-selected recipe uses six age states (including unknown).

2.3 C1: pathology-specific conditioning

Each general pathology query has a conditioned twin. The conditioned query attends to H_C through a gated cross-attention residual and is then modulated by FiLM:

$$\tilde{q}_c = q_c + g_c \text{CrossAttn}(q_c, H_C), \quad (2)$$

$$q_c^{\text{ind}} = \gamma_c(H_C) \odot \tilde{q}_c + \beta_c(H_C). \quad (3)$$

The modulation is initialized near identity ($\gamma \approx 1$, $\beta \approx 0$), allowing the pretrained CT pathway to remain stable early in adaptation. A clinical/global conditioned slot is added alongside the C pathology twins. The resulting bank contains $13 + 2C$ slots: 49 for the 18-class adult task and 59 for the 23-class pediatric task.

2.4 C2: class-specific clinical relevance

For each pathology c , C2 compares the indication embedding e_I with a frozen class-prompt embedding p_c :

$$r_c = \sigma(\text{MLP}[e_I; p_c; e_I \odot p_c]), \quad (4)$$

$$w_c = 1 + \text{softplus}(b)r_c. \quad (5)$$

The non-negative weight w_c controls conditioned-pathology pooling and the conditioned prompt loss. The relevance bias is initialized at $b = -2$, so C2 starts as a modest correction rather than suppressing incidental findings.

2.5 Primary fusion and readouts

The primary model pools the C conditioned pathology twins with C2 weights and averages that representation with the clinical/global slot:

$$z_{\text{pool}} = \frac{\sum_c w_c t_c^{\text{ind}}}{\sum_c w_c}, \quad z_{\text{ind}} = \frac{1}{2}(z_{\text{pool}} + z_{\text{clin}}), \quad (6)$$

$$z_{\text{final}} = W_f[z_{\text{gen}}; z_{\text{ind}}], \quad W_f \leftarrow [I \ 0]. \quad (7)$$

Both branches use the same positive/negative pathology-prompt softmax. The general branch reads z_{gen} and the primary conditioned branch reads z_{final} . A separate registered `class_logit` ablation instead applies the pathology-specific residual correction

$$\ell_c = \ell_c^{\text{gen}} + \alpha_c \Delta \ell_c^{\text{ind}}, \quad (8)$$

where α_c is learned and initialized so that the model begins close to the CT-only parent. This ablation is not presented as the primary fusion.

We also evaluate a supervised auxiliary readout,

$$\ell^{\text{cls}} = W \text{LN}([z_{\text{gen}}; z_{\text{ind}}]) + b, \quad (9)$$

which is a LayerNorm followed by one linear multilabel head. This `ctx_cls` result is always labeled supervised and is not conflated with prompt-based inference.

2.6 Primary training objective

The transferred prompt recipe optimizes

$$\mathcal{L} = \mathcal{L}_{\text{clip}} + 0.5\mathcal{L}_{\text{prompt}} + 0.5\mathcal{L}_{\text{align}} + 0.5\mathcal{L}_{\text{ptok}} + 2.0\mathcal{L}_{\text{ind}}. \quad (10)$$

The primary prompt arm uses temperature one, null/empty negative prompts, no L2 normalization, and checkpoint selection by conditioned development AUROC. The counterfactual loss and auxiliary `ctx_cls` loss are zero in the primary arm and are introduced only in named ablations.

3 Experimental Design

3.1 Cohorts and splits

Table 1 summarizes the controlled evaluation cohorts. The historical age ≥ 5 pediatric task excludes examinations below age five and removes four adult-specific targets from both training and evaluation. The removed targets are arterial wall calcification, coronary wall calcification, hiatal hernia, and emphysema. The frozen paper campaign retains all BCH examinations with $0 \leq \text{age} < 18$ while using the same 23 pediatric-relevant targets. No CT-RATE scans are mixed into pediatric adaptation minibatches. Adult knowledge enters only through the ARC-CT initialization lineage.

3.2 Evaluation

The primary endpoint is macro AUROC across trained targets. We report three independent seeds for matched CT-only and Refined-C1 experiments. Pediatric age strata include only classes with at least 10 positives and 10 negatives. Non-AUROC benchmark metrics follow the CT-CLIP/ARC-CT convention: per-class ROC-vertex thresholds, positive-class precision, accuracy, and support-weighted F1, averaged over classes. We preserve fixed-threshold analyses separately and do not mix their metrics with this protocol.

Patient-specific context usage is tested without retraining by evaluating each Refined-C1 checkpoint with (i) the correct indication, (ii) all indications blanked, and (iii) indications shuffled across patients. This intervention isolates semantic context use from extra model capacity.

3.3 Split audit and frozen pediatric paper protocol

An MRN-level audit found no patient overlap between the historical clean BCH training and validation pools. It also found that the previously described 1,507-case “final” cohort was not an independent test set: all 1,507 scored cases were contained in the 1,509-case Stage-2 model-selection validation list. We therefore relabel those results as historical held-out validation and do not describe them as sealed test performance.

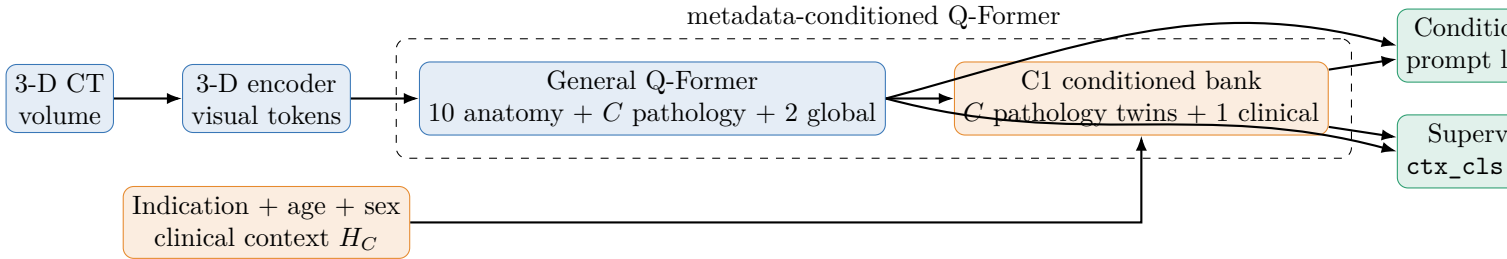


Figure 1: Architecture overview. The metadata-independent general bank is retained, while pathology-specific conditioned twins attend to clinical context through C1 cross-attention and FiLM. Prompt inference is the primary novelty readout. `ctx_cls` is a separate supervised auxiliary head.

Table 1: Datasets and matched experimental splits.

Task	Train	Development	Final validation	Classes	Provenance
Historical pediatric CT-only	5,948	1,509	—	23	BCH age ≥ 5 ; patient-disjoint train/validation; the 1,507 scored cases are a subset of model-selection validation
Historical pediatric Refined C1	5,948	1,509	—	23	Same cohort/split; seed-matched CT-only parent; base frozen, C1 trained
New pediatric < 18 protocol	4,320	748	1,273	23	MRN-disjoint train/development/test; includes ages 0–5; frozen campaign launched September 11, 2026
Adult CT-only	42,544	4,589	3,002	18	CT-RATE only; official validation split
Adult Refined C1	42,544	4,589	3,002	18	Same cohort/split; indication, scalar age, and sex conditioning

For the frozen paper campaign, we retain every study with $0 \leq \text{age} < 18$, including ages 0–5, and use the same genuine 23-target schema. The historical training pool was divided by MRN into 4,320 training studies (2,108 patients) and 748 development studies (373 patients). The historical patient-disjoint validation pool contributes 1,273 test studies (611 patients). Patient overlap is zero for every pair of splits. Because this source validation pool was inspected during earlier exploratory work, we call this a frozen retrospective test rather than claiming a never-observed prospective test.

The registered 35-arm component matrix changes one factor at a time around the full recipe: CT-only, C1 off, C2 off, cross-attention only, FiLM only, indication only, demographics only, no metadata, naive pooled-metadata fusion, anatomy queries off, routing off, regional alignment off, fully mask-free operation, scalar versus banded age, indication dropout 0/0.1/0.3/0.4, augmentation off, counterfactual loss, Qwen versus keyword region text, and the auxiliary `ctx_cls` head. It additionally tests L2 normalization, standard versus null negative prompts, indication-loss weights 1/2/4, one/two/six global queries, counterfactual weights 0/0.05/0.15, and general-versus-conditioned checkpoint selection. Numerical cells will be populated only from completed artifacts matched to their checkpoint,

split, and readout.

4 Results

4.1 Frozen true-pediatric (< 18) results

Results are pending for the registered jobs in Table 2. No numerical placeholders are treated as evidence. The registered all-age-versus-age ≥ 5 training factorial is pending.

4.2 Completed historical age ≥ 5 current-recipe ablations

4.3 Interim recovery results from pre-served checkpoints

The September 11 training wave was interrupted by a storage-quota error after its first one or two validation boundaries. Table 5 reports frozen-test point estimates from the atomically saved checkpoints while training resumes. These values are useful early evidence, but they are not substitutes for the preregistered completed-training comparison and confidence intervals. In particular, the age ≥ 5 CT-only seed 2 checkpoint was not written before the interruption.

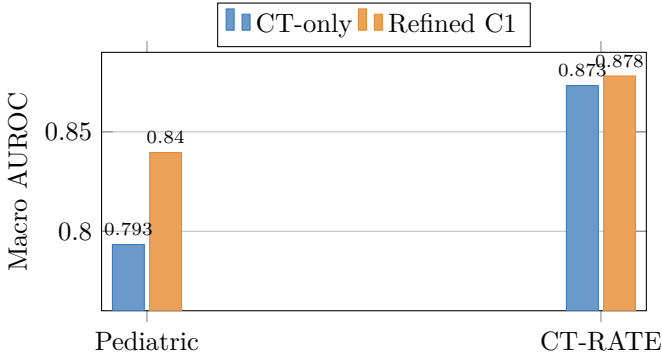


Figure 2: Refined C1 improves both matched tasks, with a substantially larger gain in the pediatric cohort.

Patient-level 1,000-replicate bootstrap intervals for the all-age conditioned AUROC are [0.8070, 0.8333], [0.7933, 0.8229], and [0.7838, 0.8133] for seeds 0–2. The corresponding age- ≥ 5 intervals are [0.8149, 0.8420], [0.8013, 0.8310], and [0.7958, 0.8271]. For the two available age- ≥ 5 CT-only controls they are [0.7691, 0.8041] and [0.7512, 0.7860]. Mean AUPRC is 0.4380 for the all-age conditioned checkpoints, 0.4442 for the age- ≥ 5 conditioned checkpoints, and 0.3760 for the two age- ≥ 5 CT-only controls.

4.4 20:00 true-pediatric checkpoint snapshot

4.5 20:00 checkpoint snapshot: does excluding ages 0–5 help?

Interpretation of the 20:00 checkpoint snapshot.

Across three seeds, metadata-conditioned prompt inference improves frozen-test macro AUROC from 0.7687 to 0.8268 ($\Delta = +0.0581$, patient-bootstrap 95% CI [0.0482, 0.0685], $p < 0.001$) and macro AUPRC from 0.3592 to 0.4616 ($\Delta = +0.1024$, 95% CI [0.0830, 0.1198]). The AUROC gain is positive in every prespecified age bin and in both sex strata. In the single-seed component snapshot, removing C1 lowers AUROC from 0.8173 to 0.7629, whereas removing C2 yields 0.8126. This interim evidence identifies token-level C1 conditioning as the larger contributor, while the converged matrix remains necessary for a final component claim. Training only on ages 5–< 18 does not improve the Full model on the identical 5–< 18 test cases ($\Delta = -0.0075$, 95% CI [−0.0136, −0.0008]), arguing against removal of younger children from the primary pediatric cohort. Finally, replacing R(2+1)D-18 with R3D-18 changes the matched seed-0 AUROC from 0.8173 to 0.8144, indicating little backbone sensitivity at this checkpoint.

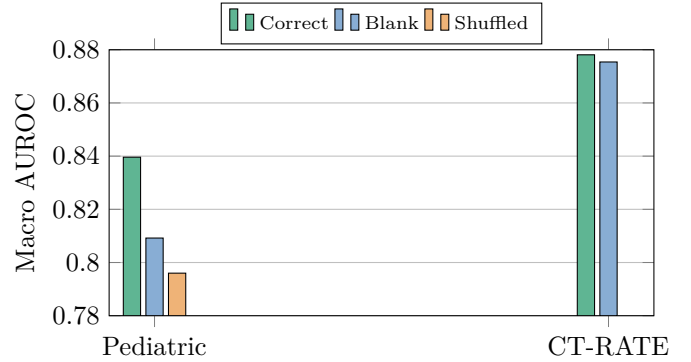


Figure 3: Inference-time indication intervention on the same trained checkpoints. Pediatric predictions depend strongly on the correct indication. The adult dependence is much smaller. The adult shuffled arm is not yet available on the final checkpoint, so no adult bar is drawn for it.

4.6 Matched pediatric and adult performance

All three pediatric C1 seeds exceed 0.83 and improve over their matched controls (Table 13). The gain is also positive in every reported age stratum (Table 14).

4.7 Does the model use the correct indication?

In pediatrics, correct indications outperform blank and shuffled indications by 3.04 and 4.36 AUROC points, respectively. Shuffling removes almost the entire gain over CT-only. In adults, the corresponding differences are only 0.20 and 0.31 points (Fig. 3). Thus, rich pediatric indications provide patient-specific diagnostic information, whereas much of the adult gain reflects the learned class-specific adaptation rather than indication semantics alone.

4.8 How predictive is metadata without the CT?

To measure clinical-prior strength and potential indication leakage directly, we freeze the same CXR-BERT indication encoder and fit only a LayerNorm and linear 23-label head, with no CT input. Table 15 reports three-seed results on the new < 18 test split. Indication alone is strongly predictive, whereas age and sex alone are near chance. Adding demographics to the indication gives only a small additional gain. Consequently, the full multimodal model must outperform both the matched CT-only model and the metadata-only control. Beating CT-only alone is insufficient evidence of visual-clinical fusion.

A conservative case-insensitive terminology audit found that at least one positive target was explicitly named in the indication for 1,064/4,320 (24.6%) training studies, 169/748 (22.6%) development studies, and 330/1,273 (25.9%) test studies. This heuristic does not distinguish

a suspected disease from a confirmed diagnosis and therefore is not itself proof of leakage. We will report model performance separately for explicitly mentioned and non-mentioned positive targets, alongside true/blank/shuffled indication interventions, to determine whether gains extend beyond direct lexical cues.

4.9 Pediatric benchmark context

Because no published pediatric benchmark uses this exact BCH cohort and 23-label task, we evaluate comparison methods locally (Table 16). These rows are not published pediatric results. CT-CLIP, CT-CLIP VocabFine, H-LIP, and BiomedCLIP are evaluated from released weights without pediatric training. The MPS-CT and GreenRFM rows are local pediatric reproductions/adaptations, and the BiomedCLIP and Merlin probe rows fit a linear head on frozen features of the pediatric training split. For methods with multiple saved checkpoints, selection is by validation macro AUROC under the same best-of- N principle used for our models: MPS-CT peaks at update 3,000 (0.6961), while the GreenRFM supervised adaptation peaks at epoch 9/10 (0.7477). We note that this selection rule changed both baselines by at most 0.001 relative to their final checkpoints, so the earlier last-checkpoint figures were not materially understated.

Several qualifications prevent this table from being read as a like-for-like architecture ranking. The prompt methods do not receive identical inputs: our method additionally receives patient metadata. The GreenRFM shared-head and the two probe rows are supervised readouts. Merlin is an abdominal model and BiomedCLIP is a 2-D model applied slice-wise, so both are reference points rather than matched volumetric CT baselines. The clean causal comparison remains Refined C1 versus its matched CT-only control.

One comparison nevertheless deserves comment. A linear probe on frozen BiomedCLIP features (0.7643) exceeds every purpose-built volumetric CT foundation model we could evaluate on this cohort, including the GreenRFM supervised adaptation (0.7477) and MPS-CT (0.6961), despite BiomedCLIP having no three-dimensional context whatsoever. Its per-class profile makes the mechanism explicit: it is competitive on findings legible in a single axial section and at chance on lung nodules. We read this as evidence that a substantial part of current volumetric pretraining benefit on pediatric CT is recoverable from a general-purpose 2-D biomedical encoder, and as a reason to report pediatric baselines by measurement rather than by inheriting adult CT-RATE rankings.

4.10 Pediatric report→CT retrieval

On the same 1,507-study pediatric cohort we also measured report→image Recall@K and image→image label-Jaccard@K for every reproducible method, with no retrieval-specific fitting. The table is generated from the

audited prediction artifacts. The complete model, Refined C1 reading out $\text{norm}(z_{\text{gen}} + z_{\text{ind}})$, leads every prior method on all ten columns. It reaches 9.5/15.5/39.9/51.7 on report→image and 32.2/30.9/27.7 on clean image→image Jaccard. The cross-model table lists only the full model and its matched CT-only control. Table 18 separates our readouts. There the metadata-free z_{gen} (shared by CT-only and Refined C1, whose base is frozen) retrieves the exact report best, at R@5/10/50/100 of 10.8/17.1/41.2/54.2, in the range of the published adult CT-RATE values, while the conditioned z_{ind} retrieves the most label-similar studies best (33.4 Jaccard at $K=5$): indication text sharpens what a study is a case of and blurs which report was dictated for it. MPS-CT is the strongest prior method (6.3/10.7/29.1/42.6), then H-LIP (3.6), BiomedCLIP (1.9) and Merlin (1.0). Released CT-CLIP and pediatric-adapted GreenRFM are at chance for report→image, GreenRFM because its unnormalized fixed-temperature contrastive space does not support cosine retrieval, the same defect that voids its prompt readout. The table reports image→image both with the query excluded from its gallery and in the CT-CLIP-script convention that prior tables (including Table 4 of the ARC-CT paper) inherit, which includes the query and accumulates across K . The two differ by roughly 20–40 points on identical latents. Text context length is disclosed rather than equalised: at their published lengths, Merlin’s 1,024 tokens truncate 5 of the 1,507 reports and CT-CLIP’s and H-LIP’s 512 truncate about 107, while the 128 that GreenRFM and MPS-CT train with truncate 1,368 (91%) against a 267-token median, bounding their recall but not their image→image scores.

4.11 Adult CT-RATE benchmark

The adult prompt model reaches 0.8781 macro AUROC on the full official 3,002-study split on one final checkpoint (Table 20). Its image-only branch reaches 0.8733 on the same forward pass. A separate augmented Biltir implementation with a direct supervised `ctx_cls` head reaches 0.8839. Because it is a supervised readout and currently a single selected checkpoint, we report it separately from the three-seed prompt result.

The newer adult-only Biltir prompt-selected checkpoints are summarized in Table 19. These single-checkpoint evaluations are kept separate from the historical local three-seed comparison because their training recipe and checkpoint family differ. In particular, jobs 4614 and 4616 retained a counterfactual-loss weight of 0.05. The clean primary rerun sets this weight to zero. The conditioned branch improves on the general branch in each paired row, but the indication-specific increment is small and the full-split no-indication result for checkpoint 4614 is higher than its true-indication result. A final true/blank/shuffled intervention is therefore still required for a clean semantic-use claim on the adult cohort.

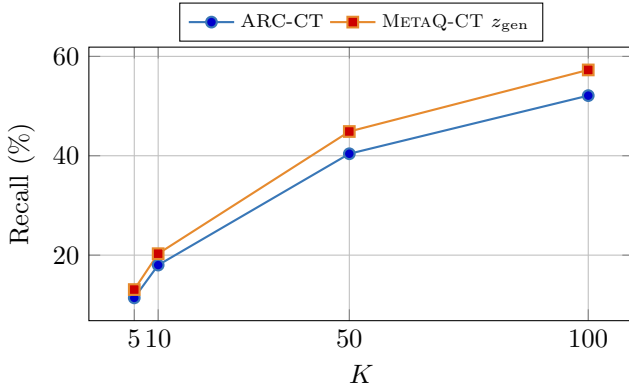


Figure 4: Canonical report-to-CT retrieval without retrieval-specific fitting.

4.12 External RAD-ChestCT transfer

Strict direct transfer does not yet lead the RAD-ChestCT benchmark (Table 21). Fitting a logistic readout on the external `imgtrain` split reaches 0.7946 AUROC on held-out data, but this is a domain-adaptation result rather than zero-shot transfer.

4.13 Report-to-CT retrieval

Without retrieval-specific fine-tuning, the canonical general representation outperforms published report-to-CT recall at every cutoff (Table 22). Mixing the metadata-conditioned global embedding does not improve retrieval. This result therefore establishes the strength of the inherited shared representation, not a retrieval gain caused by C1.

4.14 Preliminary report generation

We attach a four-layer resampler and matched Qwen3-8B LoRA decoder to either 30 general tokens (CT-only) or all 49 adult C1 tokens. Both arms map to exactly 32 prefix tokens and optimize report teacher forcing plus an identical class-balanced 18-label auxiliary loss. Metadata text is never supplied directly to the language model.

The CT-only arm completed generation on 4,589 development studies, reaching BLEU-1/2/3/4 of 0.4092/0.3202/0.2665/0.2309 and ROUGE-L of 0.4236. Its best development loss is 0.53985. The recoverable C1 checkpoint is effectively tied at 0.54004. We do not claim a C1 report-generation gain until correct, blank, and shuffled-indication generations, F1-RadGraph, RadFact-CT, bootstrap intervals, and blinded radiologist review are complete. These preliminary values are intentionally excluded from the abstract’s claims.

5 Discussion

The pediatric improvement is large, seed-consistent, positive across all 23 pathologies, and nearly eliminated by

assigning the wrong indication. This pattern supports a clinically plausible mechanism: detailed pediatric indications provide pathology-specific prior information that is aligned with the conditioned query design. The largest gains occur for pneumothorax, pulmonary metastases, post-treatment change, pulmonary cysts, and mosaic attenuation.

The adult result tells a different story. Correct, blank, and shuffled indications yield 0.8781 and 0.8754 AUROC for the true and blanked arms on the final checkpoint, with the shuffled arm not yet available. The full architecture improves over CT-only, but the small intervention gap indicates that adult CT-RATE indications are often missing or generic and that much of the improvement arises from class-specific residual adaptation. We therefore avoid attributing the entire adult gain to indication semantics.

Retrieval further clarifies which component contributes. The isolated general bank retains a strong canonical embedding and achieves leading recall, whereas post-hoc metadata mixing slightly degrades it. Metadata conditioning was optimized for class-specific diagnosis rather than global report alignment. A retrieval-specific conditioned contrastive objective is needed before claiming metadata-aware retrieval improvement.

6 Limitations

The pediatric cohort is retrospective and single-institutional. The historical 1,507-case score was obtained on the Stage-2 model-selection validation cohort, not an independent final test, and is therefore labeled validation throughout. The new MRN-disjoint < 18 protocol reduces this ambiguity, but its source test pool was visible in earlier exploratory analyses. The age- ≥ 5 experiment changes both cohort and target schema relative to earlier all-age work, so it does not establish that younger children corrupt training. A factorial all-age versus age- ≥ 5 comparison with the same 23 labels is required. Every pediatric comparison row is a local measurement rather than a published external benchmark, and several are not architecture-matched: the BiomedCLIP rows apply a 2-D encoder slice-wise with no volumetric context, the Merlin row transfers an abdominal model, and the two probe rows use supervised linear heads rather than prompt inference. Four further methods reported on adult CT-RATE (BIUD, ViSD-Boost, BrgSA, and fVLM) could not be measured at all because their releases lack code, weights, training entry points, or a license. We state this rather than importing their adult figures into a pediatric table. A pediatric CT-CLIP fine-tune also remains outstanding, since its preprocessing cache exceeded available scratch quota. Only its zero-shot transfer is reported. The strongest supervised CT-RATE result comes from a separate single-checkpoint implementation and needs multi-seed replication. Direct RAD-ChestCT transfer remains below the strongest published baselines. Finally, report-generation clinical metrics and radiologist review are not yet com-

plete.

7 Conclusion

On historical held-out validation, pathology-specific conditioning of a volumetric CT Q-Former with indication, age, and sex produces a reproducible 4.63-point pediatric macro-AUROC gain and a 2.09-point adult CT-RATE gain over matched CT-only models. Counterfactual inference shows that the pediatric improvement depends strongly on the correct clinical indication. The model also preserves a strong general representation for cross-modal retrieval. Together, these findings motivate clinical context as an explicit, auditable component of volumetric medical vision-language models.

A Complete Per-Class Results

B Auxiliary Supervised Readout

For completeness, the best separate Biltir `ctx_cls` checkpoint reaches 0.88386 macro AUROC on all 3,002 CT-RATE validation studies and 0.88484 on the 1,464-study indication-present subset. Full-split per-class AUROCs are listed in Table 25. This appendix is deliberately separate from the prompt-based primary result.

References

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- [4] Yingtai Li, Shuai Ming, Qiuli Wang, Mingyue Zhao, Hongchun Zhang, Yuhe Tian, Haoran Lai, Rongsheng Wang, Rui Zhou, Rundong Wang, Yujia Li, Zhiyang He, Xiaodong Tao, Wei Chen, Wei Wei, and Shaohua Kevin Zhou. GreenRFM: Learning a resource-efficient radiology vision-language foundation model via supervision-centric pre-training, 2026.

Table 2: Live pediatric paper experiment ledger. A dash means that no result is yet available. Queued jobs are not completed evidence.

Track	Scheduler ID	Deliverables at September 18
Historical age- $\geq$ 5 completion	23949615	Mississippi C1/C2, meta-data, anatomy, routing, alignment, mask, prompt, and training-recipe arms
Historical age- $\geq$ 5 evaluation	23949616, 23949617	Consistency prompt, per-class, and age-stratified outputs
Historical age- $\geq$ 5 table render	23951526	Reflex partial table generation every completed transferred-recipe arm in the imported L ^A T _E X table
True-pediatric smoke tests	23949882, 23949883	Stage 1 Completed successfully plus full, cross-attention-only, FiLM-only, and naive-fusion startup checks
True-pediatric Stage-1	23949884	R(2+1) Completed; best dev 18 adap-tation

Table 3: Completed current-recipe age- ≥ 5 23-target ablations on the same historical validation split. Global denotes the conditioned/fused prompt readout; arms without a completed evaluation are omitted.

Arm	Global AUC	General AUC	Global F1	Global accuracy	Global precision
Full model, three-seed mean	0.8387	0.8143	0.3644	0.8675	0.5082
CT-only, three-seed mean	0.8050	0.8050	0.3303	0.8729	0.5174
Full current model, seed 0	0.8370	0.8076	0.3134	0.8695	0.5334
Full current model, seed 1	0.8393	0.8167	0.3400	0.8745	0.4924
Full current model, seed 2	0.8399	0.8185	0.4397	0.8586	0.4987
CT-only, seed 0	0.8177	0.8177	0.3738	0.8776	0.5820
CT-only, seed 1	0.8159	0.8159	0.3514	0.8792	0.5517
CT-only, seed 2	0.7813	0.7813	0.2657	0.8617	0.4185
Indication dropout 0.4	0.8216	0.7882	0.3314	0.8512	0.4987
C1 off	0.8103	0.7893	0.2896	0.8016	0.3982
Indication conditioning only	0.8204	0.7883	0.3591	0.8152	0.3997
Demographic conditioning only	0.8085	0.7890	0.3874	0.8298	0.4060
Standard negative prompts	0.8201	0.7767	0.3348	0.8257	0.4523
Auxiliary ctx_cls	0.8243	0.7880	0.3781	0.8391	0.4762
Indication-loss weight 1	0.8200	0.7862	0.3520	0.8173	0.4350
Indication-loss weight 4	0.8229	0.7893	0.3634	0.8359	0.4199
Qwen region text	0.8206	0.7877	0.4036	0.7958	0.3817
One global query	0.8221	0.7879	0.3274	0.8478	0.3971
Six global queries	0.8238	0.7940	0.3294	0.8044	0.4276
Augmentation off	0.8223	0.7795	0.3587	0.8039	0.3523

Table 4: Available current-recipe indication interventions on the age- ≥ 5 cohort. Only completed evaluations are shown.

Seed	Indication	AUROC
0	Correct	0.8370
0	Blank	0.8180
1	Correct	0.8393
1	Blank	0.8204
1	Shuffled	0.8107
2	Correct	0.8399

Table 5: Interim 23-label pediatric frozen-test macro AUROC from preserved checkpoints. Full denotes the metadata-conditioned prompt branch; general is the metadata-independent branch from the same full-model checkpoint.

Cohort / architecture	Readout	Seed 0	Seed 1	Seed 2	Mean	Qualification
All age < 18, R(2+1)D-18	Full conditioned	0.8210	0.8086	0.7994	0.8097	updates 500–1000
All age < 18, R(2+1)D-18	General branch	0.7903	0.7726	0.7701	0.7777	same checkpoints
All age < 18, R(2+1)D-18	Within-checkpoint Δ	+0.0307	+0.0360	+0.0293	+0.0320	conditioned–general
Age $\geq$ 5, R(2+1)D-18	Full conditioned	0.8295	0.8165	0.8121	0.8194	updates 500–1000
Age $\geq$ 5, R(2+1)D-18	CT-only	0.7873	0.7693	—	0.7783	two seeds available
Age $\geq$ 5, R(2+1)D-18	Matched Δ	+0.0422	+0.0472	—	+0.0447	paired two-seed mean
All age < 18, R3D-18	Full conditioned	0.8044	—	—	0.8044	update 1000

Table 6: Interim 20:00 checkpoint snapshot on the MRN-disjoint < 18 test split; training continued after evaluation.

Seed	CT-only AUROC	Full AUROC	Δ
0	0.7719	0.8173	+0.0454
1	0.7687	0.8313	+0.0625
2	0.7654	0.8317	+0.0662
Mean	0.7687	0.8268	+0.0581

Patient-bootstrap 95% CI for the mean paired AUROC difference: [0.0482, 0.0685]; two-sided paired bootstrap $p < 0.001$.

Table 7: Audited completed < 18 checkpoint-snapshot and follow-up ablations. Arms without a completed frozen-test evaluation are omitted.

Arm	AUROC	AUPRC	Brier	ECE	Sens.@95% spec.
Full model, seed 0	0.8173	0.4619	0.1155	0.1093	0.4307
Full model, seed 1	0.8313	0.4547	0.1430	0.1546	0.4316
Full model, seed 2	0.8317	0.4681	0.1069	0.0941	0.4504
CT-only, seed 0	0.7719	0.3547	0.1012	0.0794	0.3321
CT-only, seed 1	0.7687	0.3681	0.0959	0.0637	0.3303
CT-only, seed 2	0.7654	0.3547	0.1025	0.0802	0.3263
C2 off	0.8126	0.4191	0.1119	0.1059	0.3842
C1 off	0.7629	0.3669	0.1292	0.1137	0.3534
R3D-18 backbone, full model, seed 0	0.8144	0.4549	0.1010	0.0879	0.4227
Asymmetric loss (ASL), seed 0	0.8055	0.4493	0.1016	0.0991	0.4017
BCE + BNM ($\lambda = 0.001$), seed 0	0.8305	0.4715	0.0989	0.0880	0.4430
BCE + BNM ($\lambda = 0.003$), seed 0	0.8301	0.4693	0.1000	0.0860	0.4370
BCE + BNM ($\lambda = 0.010$), seed 0	0.8298	0.4676	0.0955	0.0813	0.4292
ASL + BNM ($\lambda = 0.001$), seed 0	0.8190	0.4558	0.1882	0.2591	0.4118
ASL + BNM ($\lambda = 0.003$), seed 0	0.8172	0.4528	0.1614	0.2271	0.3998
ASL + BNM ($\lambda = 0.010$), seed 0	0.8158	0.4462	0.2103	0.2832	0.4110

Table 8: Age and sex subgroup consistency on the < 18 test split.

Subgroup	Model	N	AUROC
age_0_2	ct_only	105	0.7185
age_0_2	full	105	0.7609
age_2_5	ct_only	147	0.7509
age_2_5	full	147	0.8051
age_5_10	ct_only	252	0.7200
age_5_10	full	252	0.7807
age_10_15	ct_only	435	0.7825
age_10_15	full	435	0.8478
age_15_18	ct_only	334	0.7427
age_15_18	full	334	0.8066
sex_F	ct_only	681	0.7715
sex_F	full	681	0.8315
sex_M	ct_only	592	0.7603
sex_M	full	592	0.8179
exam_year_2011_2018	ct_only	615	0.7809
exam_year_2011_2018	full	615	0.8427
exam_year_2019_2026	ct_only	658	0.7587
exam_year_2019_2026	full	658	0.8090
acquisition_site_rank_1	ct_only	413	0.7516
acquisition_site_rank_1	full	413	0.8057
acquisition_site_rank_2	ct_only	293	0.7089
acquisition_site_rank_2	full	293	0.7816
acquisition_site_rank_3	ct_only	135	0.7164
acquisition_site_rank_3	full	135	0.7792
acquisition_site_rank_4	ct_only	116	0.7078
acquisition_site_rank_4	full	116	0.7649

Table 9: Performance stratified by explicit target terminology in the indication.

Terminology	Model	AUROC	AUPRC
explicit_term_present	ct_only	0.6694	0.6771
explicit_term_present	full	0.7153	0.7339
explicit_term_absent	ct_only	0.7692	0.3494
explicit_term_absent	full	0.8249	0.4486

Table 10: Complete three-seed per-class results on the frozen < 18 test split.

Class	Pos.	CT AUC	Full AUC	Δ AUC	CT AUPRC	Full AUPRC	Δ AUPRC
Medical material	535	0.8941	0.9225	+0.0283	0.8649	0.9061	+0.0412
Cardiomegaly	29	0.9185	0.9609	+0.0424	0.2857	0.4413	+0.1556
Pericardial effusion	44	0.8471	0.9024	+0.0553	0.1890	0.2753	+0.0862
Lymphadenopathy	96	0.7745	0.7602	-0.0142	0.2616	0.2732	+0.0115
Atelectasis	447	0.7270	0.7465	+0.0195	0.6210	0.6560	+0.0350
Lung nodule	631	0.6579	0.6891	+0.0312	0.6223	0.6656	+0.0433
Lung opacity	426	0.7438	0.7741	+0.0303	0.6080	0.6536	+0.0456
Pulmonary fibrotic sequela	235	0.6614	0.7682	+0.1068	0.3138	0.4332	+0.1194
Pleural effusion	48	0.9559	0.9640	+0.0081	0.5193	0.6318	+0.1125
Mosaic attenuation pattern	185	0.7351	0.8253	+0.0901	0.3540	0.4362	+0.0823
Peribronchial thickening	168	0.7960	0.8352	+0.0392	0.4607	0.5047	+0.0439
Consolidation	76	0.8632	0.8922	+0.0290	0.4035	0.4380	+0.0346
Bronchiectasis	103	0.9004	0.9337	+0.0334	0.5713	0.7101	+0.1388
Interlobular septal thickening	76	0.8444	0.8818	+0.0374	0.3224	0.4314	+0.1090
Post-surgical or post-treatment change	392	0.6754	0.8150	+0.1396	0.4952	0.6828	+0.1876
Pulmonary metastases	76	0.6155	0.8170	+0.2014	0.1397	0.2708	+0.1311
Tree-in-bud	90	0.7893	0.8602	+0.0709	0.3177	0.4501	+0.1324
Pulmonary cyst	50	0.6645	0.7251	+0.0606	0.0690	0.1199	+0.0509
Mass or neoplasm	33	0.7224	0.8218	+0.0994	0.1285	0.2556	+0.1271
Mucus plugging	44	0.7891	0.8378	+0.0487	0.1760	0.2986	+0.1226
Pleural thickening or nodule	146	0.6757	0.6950	+0.0193	0.2283	0.2630	+0.0347
Bone lesion or fracture	179	0.5925	0.6699	+0.0774	0.2016	0.3056	+0.1040
Pneumothorax	25	0.8363	0.9176	+0.0814	0.1072	0.5136	+0.4065

Table 11: Interim 20:00 age-training checkpoint snapshot evaluated on the identical 5-< 18 MRN-disjoint test subset.

Model	Training ages	AUROC	AUPRC
CT-only	0-< 18	0.7656	0.3573
CT-only	5-< 18	0.7907	0.4110
Full	0-< 18	0.8287	0.4689
Full	5-< 18	0.8212	0.4503

CT-only: excluding ages 0-5 changes AUROC by +0.0250 (patient-bootstrap 95% CI [0.0154, 0.0344]).

Full: excluding ages 0-5 changes AUROC by -0.0075 (patient-bootstrap 95% CI [-0.0136, -0.0008]).

Table 13: Pediatric seed and indication intervention results.

Arm	Seed 0	Seed 1	Seed 2	Mean
CT-only	0.7934	0.7950	0.7916	0.7933
C1, correct	0.8386	0.8414	0.8389	0.8396
C1, blank	0.8111	0.8091	0.8074	0.8092
C1, shuffled	0.7980	0.7967	0.7934	0.7960

Table 12: Historical matched prompt-inference validation results (macro AUROC, mean $\pm$ SD over three seeds). These are not independent test estimates.

Task	CT-only	Refined C1	All $\geq$ 18
Pediatric, 23 labels	0.7933 $\pm$.0017	0.8396$\pm$.0015	0.8463$\pm$.0015
CT-RATE, 18 labels (final checkpoint)	0.8733	0.8781	0.8781

Table 14: Pediatric performance by age. Only eligible classes enter each stratum's macro AUROC.

Age	N	Classes	CT-only	C1 (Δ)
5-10	252	15	0.7451	0.7904 (+.0452)
10-18	768	23	0.7995	0.8440 (+.0445)
18-25	402	20	0.7729	0.8190 (+.0461)
25+	85	15	0.7615	0.8050 (+.0435)
Refined C1	1,020	23	0.7979	0.8412 (+.0433)
All $\geq$ 18	487	23	0.7841	0.8325 (+.0483)
Pooled	1,507	23	0.7933	0.8396 (+.0463)

Table 15: No-CT controls on the frozen < 18 test split ($N = 1,273$). AUROC is mean $\pm$ SD over three linear-head seeds. AUPRC is the three-seed mean.

Input (no CT)	Macro AUROC	Macro AUPRC
Age + sex	0.5064 ± 0.0065	0.1578
Indication	0.7860 ± 0.0003	0.3792
Indication + age + sex	0.7867 ± 0.0010	0.3801

Table 16: Pediatric benchmark evaluation on the same 1,507 studies (648 patients) and 23 labels. All values are locally measured macro AUROC, not published pediatric results. Rows are grouped by whether the method was adapted to the pediatric cohort, because a pediatric-adapted model and an untrained transfer answer different questions and should not be ranked in a single list. Intervals are 1,000-sample bootstrap resampling *patients*, not studies. The cohort averages 2.33 studies per child, and resampling studies understates the interval by roughly 35%.

Method	Readout	AUROC	95% CI	Qualification
<i>Adapted to the pediatric cohort</i>				
METAQ-CT Refined C1	conditioned prompt	0.8396^a	—	three-seed historical campaign
Our CT-only	prompt	0.7933 ^a	—	matched three-seed control
BiomedCLIP, linear probe	supervised probe	0.7643	0.7473–0.7803	2-D encoder, 32 axial slices, mean-pooled ^b
GreenRFM, local adaptation	supervised head	0.7477	0.7310–0.7637	best of 10 saved checkpoints (epoch 9)
Merlin, linear probe	supervised probe	0.7144	0.6951–0.7313	abdominal/pelvic CT model ^c
MPS-CT, local reproduction	zero-shot prompt	0.6961	0.6840–0.7089	best of 12 saved checkpoints (update 3,000)
GreenRFM, local reproduction	zero-shot prompt	0.6620	—	prompt readout is ill-posed for this model ^d
<i>Released adult weights, no pediatric training</i>				
BiomedCLIP	zero-shot prompt	0.6938	0.6695–0.7108	fixed prompt pair, no tuning ^b
H-LIP	zero-shot prompt	0.6916	0.6763–0.7048	released CT-RATE weights
CT-CLIP	zero-shot prompt	0.5478	0.5277–0.5670	best of the four templates in the released code ^e
CT-CLIP VocabFine	zero-shot prompt	0.5339	0.5113–0.5572	second released CT-CLIP variant ^e

^a Historical Refined-C1 campaign (scalar age, 10% indication dropout), reported with three-seed standard deviations in Table 12 rather than a bootstrap interval. The remaining rows are single trained models scored with the identical metric code on the identical split.

^b BiomedCLIP is a 2-D biomedical vision–language model applied slice-wise with no inter-slice context, so findings defined by three-dimensional extent are invisible to it by construction: it reaches 0.503 (chance) on lung nodule against 0.935 on pleural effusion. It is included as a deliberately non-CT-specific reference point, not as an architecture-matched comparison. Its probe regularization was selected by patient-grouped cross-validation on the pediatric training split alone.

^c Merlin is an abdominal/pelvic CT foundation model, so this row is out-of-domain on both body region and adult-to-pediatric transfer. Its preprocessing also resamples to 3 mm slices and discards thin-slice detail.

^d GreenRFM’s contrastive objective uses unnormalized dot products at fixed temperature, so per-volume embedding norm enters the cross-volume ranking. Its published adult result is the supervised head, which is the row above. We report the prompt value only for completeness.

^e Sweeping all four prompt templates shipped in the CT-CLIP repository spans 0.528–0.548, so the low value is not a template artifact. A supervised probe fitted on CT-CLIP’s own image latents reaches only 0.609, indicating a representational rather than a readout limitation on pediatric CT.

Table 17: Pediatric cross-modal retrieval on the historical cohort (1,507 studies, 648 patients), no retrieval-specific fitting. Report→image is exact-match Recall@K over all 1,507 studies. Image→image is mean label-set Jaccard of the top- K retrieved studies (gallery = studies with ≥ 1 positive label): *clean* excludes the query from its own gallery; *Table-4 convention* reproduces the CT-CLIP evaluation script used by prior work, which includes the query and accumulates across K , and is shown only for comparability with published adult tables. Our arms are three-seed means; chance Recall@K is K/N .

Method	Report→image R@K				Image→image, clean			Image→image, Table-4 conv.		
	@5	@10	@50	@100	@5	@10	@50	@5	@10	@50
<i>Adapted to the pediatric cohort</i>										
METAQ-CT (full, metadata-conditioned) ^m	9.5±0.4	15.5±0.2	39.9±0.6	51.7±1.4	32.2±0.6	30.9±0.6	27.7±0.4	67.7	57.5	50.4
METAQ-CT / CT-only control (metadata-free)	10.8±0.6	17.1±0.1	41.2±0.4	54.2±1.2	31.7±0.2	30.4±0.3	27.5±0.1	67.4	57.2	50.1
GreenRFM, local adaptation (epoch 9)	0.3	1.1	5.4	9.4	27.8	26.2	23.4	66.0	54.9	47.4
MPS-CT, local reproduction (update 3,000)	6.3	10.7	29.1	42.6	30.9	28.4	24.2	67.3	56.5	48.8
<i>Released weights, no pediatric training</i>										
H-LIP	3.6	5.8	14.9	22.4	23.7	22.6	20.8	64.2	52.7	45.0
BiomedCLIP (2-D, slice-wise)	1.9	3.3	11.8	17.4	25.3	23.8	21.0	64.9	53.5	45.8
Merlin (abdominal)	1.0	1.7	6.2	10.9	21.8	21.3	19.9	63.4	51.7	44.1
CT-CLIP	0.9	1.3	5.4	10.2	19.8	19.4	18.8	62.6	50.6	43.0
CT-CLIP VocabFine	0.3	0.9	3.7	6.7	20.8	20.0	19.1	62.9	51.0	43.4
Chance (K/N)	0.3	0.7	3.3	6.6	–	–	–	–	–	–

^mUses each gallery study’s own indication, age, and sex through the conditioned bank; not an equal-input comparison with the metadata-free rows, shown to test whether conditioning changes retrieval. Every model encodes the identical impression-first report string; each uses its own tokenizer at its own trained context length, which truncates long reports: ours 256 tokens; CT-CLIP 512 (107 of 1,507 truncated); VocabFine 512 (107 of 1,507 truncated); H-LIP 512 (106 of 1,507 truncated); GreenRFM 128 (1,368 of 1,507 truncated, 91%); MPS-CT 128 (1,368 of 1,507 truncated, 91%); BiomedCLIP 256 (846 of 1,507 truncated, 56%); Merlin 1,024 (5 of 1,507 truncated). Single-model rows carry 1,000-draw patient-bootstrap intervals in the accompanying JSON.

Table 18: Readout ablation for METAQ-CT on the same pediatric retrieval task and the same stored latents as Table 17; three-seed means. The two tasks have opposite winners: the metadata-free readout retrieves the exact report best, the conditioned readout retrieves the most label-similar studies best.

Readout	R@5	R@100	Jaccard@5	Jaccard@50
z_{gen} , metadata-free (shared with CT-only)	10.8±0.6	54.2±1.2	31.7±0.2	27.5±0.1
$\text{norm}(z_{\text{gen}}+z_{\text{ind}})$, full model ^m	9.5±0.4	51.7±1.4	32.2±0.6	27.7±0.4
z_{ind} , conditioned only ^m	8.0±1.0	47.2±1.1	33.4±0.7	28.0±0.6

^mUses each gallery study’s own indication, age, and sex; not an equal-input comparison with the metadata-free row.

Table 19: Latest adult Biltir paired prompt evaluation (job 4689). Values are macro AUROC from the same checkpoint and evaluation subset. “None” passes the literal NO_INDICATION input.

Checkpoint / training cohort	Evaluation subset	General	True indication	None	$\Delta(\text{true-general})$
4616, full CT-RATE	Full ($N = 3,002$)	0.8735	0.8764	0.8734	+0.0029
4616, full CT-RATE	Indication present ($N = 1,464$)	0.8755	0.8764	0.8736	+0.0009
4614, indication-present only	Full ($N = 3,002$)	0.8698	0.8710	0.8725	+0.0012
4614, indication-present only	Indication present ($N = 1,464$)	0.8748	0.8820	0.8773	+0.0072

Table 20: CT-RATE 18-label benchmark under the paper operating-threshold protocol. Higher is better.

Method	AUROC	F1	Accuracy	Precision	Readout / qualification
CT-CLIP [1]	0.731	0.707	0.668	0.323	published prompt baseline
MPS-CT [3]	0.838	0.796	0.774	0.435	published zero-shot prompt result
GreenRFM [4]	0.848	0.800	0.777	0.444	published zero-shot prompt result
ARC-CT [2]	0.855	0.809	0.787	0.455	published prompt baseline
METAQ-CT Refined C1	0.869	0.817	0.796	0.471	prompt-based; earlier three-seed campaign, AUROC mean 0.8683. The newer final checkpoint reaches 0.8781 AUROC; its F1, accuracy and precision were not exported under this protocol
METAQ-CT + <code>ctx_cls</code> [†]	0.884	0.836	0.820	0.503	supervised one-layer readout; separate Biltir implementation

[†] This row is not prompt-only zero-shot inference and must not be used as the primary proof of metadata-conditioned prompt novelty.

Table 21: RAD-ChestCT results. Direct rows use no RAD-ChestCT labels.

Method	AUROC	F1	Acc.	Prec.
ARC-CT	0.734	0.716	0.679	0.392
MPS-CT	0.773	0.748	0.719	0.450
GreenRFM	0.780	0.757	0.730	0.461
METAQ-CT historical Refined C1 prompt, direct	0.749	0.731	0.696	0.404
METAQ-CT Biltir 4257 general prompt, direct	0.7618	—	—	—
METAQ-CT Biltir 3748 <code>ctx_cls</code> , direct	0.747	0.744	0.711	0.417
METAQ-CT adapted [†]	0.7946	0.7735	0.7461	0.4466

A dash indicates that the corresponding metric was not exported under the same operating-threshold protocol. [†]Logistic readout fitted on RAD-ChestCT `imgtrain` and evaluated on held-out `imgvalid+imgtest` ($N = 1,344$). Not zero shot.

Table 22: Report-to-CT retrieval on all 3,002 CT-RATE validation studies (%).

Method	R@5	R@10	R@50	R@100
CT-CLIP	2.9	5.0	18.0	28.7
MPS-CT	10.0	16.3	39.0	52.2
GreenRFM	9.5	16.0	39.9	54.3
ARC-CT	11.4	18.0	40.4	52.1
METAQ-CT z_{gen}	13.07	20.29	44.88	57.27
METAQ-CT $z_{\text{gen}} + z_{\text{ind}}$	12.91	20.28	44.37	56.53

Table 23: Pediatric age ≥ 5 per-class AUROC, averaged over three seeds.

Pathology	CT-only	C1	Δ	Pathology	CT-only	C1	Δ
Medical material	.8562	.9024	+.0463	Peribronchial thickening	.8236	.8431	+.0195
Cardiomegaly	.9441	.9528	+.0088	Consolidation	.9241	.9286	+.0045
Pericardial effusion	.7880	.8125	+.0245	Bronchiectasis	.9101	.9250	+.0149
Lymphadenopathy	.8042	.8163	+.0121	Interlobular septal thickening	.8462	.8828	+.0366
Atelectasis	.7126	.7233	+.0107	Post-surgical/treatment change	.7485	.8658	+.1174
Lung nodule	.6507	.6937	+.0431	Pulmonary metastases	.6662	.8036	+.1374
Lung opacity	.8148	.8329	+.0181	Tree-in-bud	.8280	.8737	+.0457
Pulmonary fibrotic sequela	.7512	.7764	+.0252	Pulmonary cyst	.7523	.8548	+.1025
Pleural effusion	.9729	.9754	+.0025	Mass/neoplasm	.8496	.8680	+.0184
Mosaic attenuation	.7413	.8368	+.0956	Mucus plugging	.8622	.9014	+.0392
Pleural thickening/nodule	.6607	.6819	+.0212	Bone abnormality	.5789	.6348	+.0559
Pneumothorax	.7606	.9245	+.1639	Macro mean	.7933	.8396	+.0463

Table 24: Adult CT-RATE per-class prompt AUROC, averaged over three seeds.

Pathology	CT-only	C1	Δ	Pathology	CT-only	C1	Δ
Medical material	.9159	.9447	+.0287	Lung nodule	.7270	.7383	+.0113
Arterial wall calcification	.9336	.9421	+.0085	Lung opacity	.8680	.8844	+.0164
Cardiomegaly	.9343	.9364	+.0021	Pulmonary fibrotic sequela	.7020	.7463	+.0443
Pericardial effusion	.8957	.9040	+.0083	Pleural effusion	.9768	.9725	-.0042
Coronary wall calcification	.9273	.9381	+.0109	Mosaic attenuation	.8163	.8954	+.0790
Hiatal hernia	.8159	.8209	+.0050	Peribronchial thickening	.8148	.8261	+.0113
Lymphadenopathy	.7786	.7923	+.0136	Consolidation	.9285	.9334	+.0049
Emphysema	.8113	.8251	+.0138	Bronchiectasis	.7995	.8242	+.0247
Atelectasis	.7967	.8103	+.0136	Interlobular septal thickening	.8120	.8951	+.0831
				Macro mean	.8475	.8683	+.0209

Table 25: Per-class AUROC of the supervised `ctx_cls` checkpoint.

Pathology	Full	Ind.-present	Pathology	Full	Ind.-present
Medical material	.9443	.9390	Lung nodule	.8052	.8037
Arterial wall calcification	.9426	.9423	Lung opacity	.8909	.9004
Cardiomegaly	.9442	.9374	Pulmonary fibrotic sequela	.7742	.7831
Pericardial effusion	.9287	.9398	Pleural effusion	.9767	.9719
Coronary wall calcification	.9460	.9354	Mosaic attenuation	.9135	.9149
Hiatal hernia	.8519	.8707	Peribronchial thickening	.8316	.8165
Lymphadenopathy	.7998	.7867	Consolidation	.9263	.9338
Emphysema	.8420	.8310	Bronchiectasis	.8553	.8518
Atelectasis	.8359	.8540	Interlobular septal thickening	.9005	.9147
			Macro mean	.88386	.88484